

Sustainability Challenges in Urban Air Operations

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Low Altitude Economics

Week 10

Learning Objectives

By the end of this class, you should be able to:

- **Identify** major environmental and social externalities associated with urban air operations.
- **Compare** pricing, operating restrictions, standards, mitigation, and compensation as policy responses.
- **Calculate** simple operational and life-cycle greenhouse-gas indicators using transparent boundaries.
- **Evaluate** privacy and public-acceptance measures using necessity, proportionality, and stakeholder evidence.
- **Design** measurable sustainability KPIs and assess who benefits, who pays, and who bears residual impacts.

Can an LAE Service Be Sustainable and Viable?

Imagine a proposed electric-aircraft corridor near a dense urban community.

- The service may reduce travel time or improve delivery and public-service performance.
- Electricity use, vehicle manufacturing, noise, visual effects, data collection, and infrastructure still create impacts.

“Electric” does not automatically mean zero-emission, quiet, private, equitable, or publicly accepted.

- **Quick poll (2 min):** Which issue is most likely to constrain your project: noise, privacy, greenhouse-gas emissions, energy demand, or unequal distribution of benefits?

Session 1

Externalities, Pigovian Instruments & Privacy Governance

Concept Primer: Externalities

An externality arises when an activity affects third parties and those effects are not fully reflected in the decision-maker's private costs or benefits.

- **Noise and visual effects:** annoyance, sleep disturbance, amenity, and land-use impacts.
- **Privacy and data:** excessive collection, surveillance concerns, security risk, and loss of trust.
- **Climate and energy:** operational electricity emissions, supply-chain emissions, and peak-load requirements.
- **Positive effects:** time savings, safer inspection, emergency response, knowledge, and network benefits.

Managerial lesson: First identify and measure the impact; then compare prevention, mitigation, pricing, compensation, and operating restrictions.

Applied Tool: Pricing a Noise externality

Economic Benchmark

Efficient Charge per Operation = Marginal External Noise Cost

- Marginal external cost depends on exposure, time, location, number of affected people, baseline noise, event frequency, and health or amenity effects.
- Decibels are logarithmic; “decibels above a limit” should not automatically be multiplied by flights and a fixed monetary rate.
- Practical instruments may use certified noise categories, time-of-day charges, quotas, curfews, route restrictions, or performance standards.
- Charges should encourage quieter operations without implying that every impact can simply be paid for.

Back-of-the-Envelope: Appraising Quieter Technology

Illustrative assumptions: A quieter-technology upgrade costs HK\$3M and reduces expected annual noise charges, monitoring cost, and disruption by HK\$1.2M.

Without the Upgrade

- Expected annual noise-related cost: HK\$1.5M
- Five-year undiscounted cost: HK\$7.5M

With the Upgrade

- Upfront cost: HK\$3M
- Residual annual cost: HK\$0.3M
- Five-year undiscounted total: HK\$4.5M
- **Illustrative five-year saving: HK\$3M**

Takeaway: The investment may be worthwhile under these assumptions. A complete appraisal should discount cash flows and test performance, maintenance, and demand effects.

Illustrative figures; not Hong Kong noise limits, charges, or technology quotations.

Privacy Governance for Onboard Cameras

Cameras may support navigation, inspection, security, or evidence collection, but privacy risk depends on what is captured, retained, linked, and used.

- Define a lawful and specific purpose before collecting personal data.
- Test necessity and proportionality and consider less intrusive alternatives.
- Minimize field of view, resolution, recording duration, identifiers, audio, and retention where they are not needed.
- Use access controls, encryption, logs, retention limits, deletion procedures, notices, and incident response.
- Onboard masking may be one control, but it is not universally required or sufficient by itself.

Hong Kong privacy guidance for drone cameras emphasizes avoiding excessive collection, notification, security, and responsible data handling.

Applied Tool: Privacy-Control Cost

Annual Privacy-Control Cost

Annual Cost = Fixed Governance and System Cost + $q \times$ Variable Processing Cost

- Fixed cost may include privacy-impact assessment, system design, notices, access controls, training, and vendor assurance.
- Variable cost may include secure processing, storage, communications, review, and deletion.
- The cost cannot be inferred from an unsupported percentage of flight operating cost.
- Compare controls using privacy effectiveness, safety compatibility, reliability, and total cost.

Back-of-the-Envelope: Privacy-Control Cost

Illustrative assumptions: 50,000 operations/year; fixed privacy-system and governance cost HK\$600k/year.

Lower Variable Cost

- Processing cost: HK\$4/operation
- Variable cost: HK\$200k
- Fixed cost: HK\$600k
- **Annual cost: HK\$800k**

Higher-Assurance Design

- Processing cost: HK\$8/operation
- Variable cost: HK\$400k
- Fixed cost: HK\$1M
- **Annual cost: HK\$1.4M**

Takeaway: The more expensive design is justified only if its additional controls materially improve privacy, security, or operational assurance.

Illustrative assumptions; not a compliance quotation or claim that either design is legally sufficient.

Back-of-the-Envelope: Carbon-Cost Scenario

Illustrative assumptions: 50,000 operations/year and a carbon value of HK\$1 per kg CO₂e.

40 kg CO₂e/Operation

- Emissions:
 $50,000 \times 40 = 2,000,000$ kg
- **Annual carbon cost: HK\$2M**

10 kg CO₂e/Operation

- Emissions: $50,000 \times 10 = 500,000$ kg
- **Annual carbon cost: HK\$0.5M**

Takeaway: A carbon value can support investment appraisal, but this is not a claim that CAD imposes such a tax. Emissions factors and boundaries must be sourced.

Illustrative carbon price and emissions intensities.

Session 2

Carbon Accounting, Life-Cycle Analysis & Public
Acceptance

Concept Primer: GHG Inventory and Life-Cycle Assessment

Corporate GHG scopes and product life-cycle assessment answer related but different questions.

- **Scope 1:** Direct emissions from sources owned or controlled by the reporting organization.
- **Scope 2:** Indirect emissions associated with purchased electricity, steam, heat, or cooling.
- **Scope 3:** Other indirect value-chain emissions, potentially including aircraft, batteries, infrastructure, maintenance, and end-of-life activities.
- **Life-cycle assessment:** Evaluates environmental impacts of a defined product or service across a specified life cycle and functional unit.

An electric aircraft may have zero in-flight exhaust emissions while still having Scope 2 and Scope 3 emissions. Corporate scopes should not simply be added without checking boundaries and double counting.

Operational Electricity Emissions

$\text{CO}_2\text{e per operation} = \text{kWh consumed} \times \text{electricity emissions factor}$

- Match units carefully, such as kg CO₂e/kWh.
- Include charging losses and relevant facility energy if they fall within the boundary.
- Distinguish location-based grid factors from market-based contractual claims where applicable.
- Compare modes using a consistent functional unit, such as passenger-km, tonne-km, or completed mission.

Back-of-the-Envelope: Electricity Emissions

Assumption: 50 kWh of electricity per operation, including charging losses only if stated.

Factor: 0.8 kg/kWh

- 50×0.8
- **40 kg CO₂e/operation**

Factor: 0.2 kg/kWh

- 50×0.2
- **10 kg CO₂e/operation**

Takeaway: Electricity source matters, but these figures alone do not establish whether the service outperforms another transport mode. Occupancy, distance, infrastructure, and life-cycle emissions also matter.

Illustrative factors; use current, sourced factors for an actual inventory.

Applied Tool: Life-Cycle Emissions per Functional Unit

Simplified Allocation

$$\text{Life-Cycle Emissions per Unit} = \frac{E_{\text{vehicle}} + E_{\text{battery}} + E_{\text{infrastructure}} + E_{\text{operations}}}{\text{Lifetime Functional Units}}$$

- Functional units may be passenger-km, tonne-km, or completed service missions.
- State lifetime, utilization, replacement cycles, occupancy or payload, system boundary, and allocation method.
- Scope 3 is not a universal fixed percentage of total emissions.
- Compare alternatives using the same boundary and functional unit.

Back-of-the-Envelope: Allocating Embodied Emissions

Illustrative assumptions: Embodied aircraft, battery, and infrastructure emissions allocated at 12 kg CO₂e per operation.

Higher-Carbon Electricity

- Electricity emissions: 40 kg
- Allocated embodied emissions: 12 kg
- **Total: 52 kg CO₂e/operation**

Lower-Carbon Electricity

- Electricity emissions: 10 kg
- Allocated embodied emissions: 12 kg
- **Total: 22 kg CO₂e/operation**

Takeaway: Decarbonizing electricity reduces operational emissions, while embodied emissions remain sensitive to lifetime use and replacement assumptions.

Illustrative allocation; not a comparative claim about jets or sustainable aviation fuel.

Concept Primer: Public Acceptance and Use Cases

AiRMOUR examined sustainable UAM through the lens of emergency medical services while also considering other applications.

- Acceptance may vary with perceived public benefit, urgency, safety, noise, privacy, fairness, trust, location, and available alternatives.
- Urgent medical applications may receive stronger support than some non-urgent applications, but acceptance is not automatic or perfectly divided into two groups.
- Infrastructure maintenance, surveying, and environmental monitoring may also be viewed favourably in some contexts.
- Local engagement and evidence are necessary before transferring European findings to Hong Kong.

Managerial lesson: Segment by use case and affected stakeholder, then test assumptions rather than assigning universal acceptance scores.

Session 3

Equity, Social License & In-Class Exercise

Concept Primer: Equity and Social Licence

Social licence refers to continuing acceptance and trust among affected communities and stakeholders. It complements, but does not replace, legal approval.

- Ask who receives time savings, service improvements, employment, and business opportunities.
- Ask who bears noise, privacy, safety, land-use, and fiscal costs.
- Consider affordability, accessibility, geographic coverage, consultation, complaints, and remedies.
- Acceptance may change over time and cannot be purchased by one symbolic public-service commitment.

Managerial lesson: Build legitimacy through performance, transparency, participation, mitigation, and fair distribution of benefits and costs.

Applied Tool: Funding a Public-Service Commitment

Net Contribution

$$\text{Net Contribution} = Q_C m_C + \text{Public-Service Payment} - Q_{PCP} - \text{Fixed Cost}$$

- $Q_C m_C$ is contribution from commercial services.
- Q_{PCP} is the incremental cost of the public-service commitment.
- Funding may come from commercial cross-subsidy, public procurement, availability payment, philanthropy, or another transparent source.
- The arrangement should specify eligibility, service level, opportunity cost, accountability, and long-term affordability.

Back-of-the-Envelope: Cost of a Public-Service Commitment

Assumptions: 100 daily slots; commercial contribution HK\$400/flight; public-service incremental cost HK\$100/flight.

All Commercial Slots

- 100 commercial flights
- **Contribution before fixed cost:**
HK\$40,000/day
- No specified public-service capacity

Twenty Priority Slots

- Commercial contribution:
 $80 \times 400 = \text{HK\$}32,000$
- Public-service cost:
 $20 \times 100 = \text{HK\$}2,000$
- **Net before fixed cost:**
HK\$30,000/day
- Opportunity cost relative to all-commercial case:
HK\$10,000/day

Takeaway: The commitment has a measurable cost. Whether it is justified and who should fund it require a public-value and distributional assessment.

Illustrative assumptions; no claim that the commitment secures approval or acceptance.

Applied Tool: Sustainability KPI Design

Area	Example KPI	Boundary or denominator
Noise	Population-weighted exposure	Route, period, and noise metric
Climate	kg CO ₂ e/passenger-km	Life-cycle boundary and occupancy
Privacy	Personal-data incidents	Operations and severity
Equity	Service availability by area	Population, need, and affordability
Governance	Complaints resolved on time	Defined process and deadline

- Define baseline, target, measurement method, frequency, owner, assurance, and corrective action.
- No universal CAD or investor requirement mandates the three ratios previously shown.
- A KPI tracks performance; it does not by itself prove profitability or social acceptance.

In-Class Exercise: Sustainability Agreement

Scenario: A high-volume corridor is proposed near a residential area.

- **Roles:** Operator, community representatives, regulator, public-service customer, and data-protection adviser
- **Tasks:**
 1. identify noise, privacy, climate, access, and distributional impacts;
 2. propose one prevention or mitigation measure for each material impact;
 3. estimate cost, effectiveness, beneficiary, payer, and residual impact;
 4. define five measurable KPIs and review triggers; and
 5. agree on complaints, incident, pause, and corrective-action procedures.
- **Timing:** 15 minutes plus 5-minute discussion.

Exercise Template: Mitigation Scorecard

Impact	Measure	KPI	Payer
Noise	Route, schedule, or technology	Exposure metric	To neg
Privacy	Data minimization and security	Incident and retention metric	To neg
Climate	Energy and utilization strategy	kg CO ₂ e/functional unit	To neg
Equity	Access or public-service design	Availability and affordability	To neg

Do not assume a 60 dB limit or a fixed number of free public-service flights. Define thresholds from evidence, regulation, operating context, and stakeholder agreement.

Key Takeaways

- Externalities should be measured and addressed through a mix of technology, standards, operating rules, pricing, compensation, and engagement.
- Noise cannot be priced reliably by multiplying flights by decibels above an assumed threshold.
- Privacy governance requires lawful purpose, necessity, proportionality, minimization, security, transparency, retention control, and accountability.
- Corporate GHG scopes and life-cycle assessment require clear and consistent boundaries; electric operation does not imply zero life-cycle emissions.
- Public acceptance varies across use cases and stakeholders and cannot be reduced to a universal medical-versus-commercial split.
- Equity commitments and KPIs need measurable outcomes, transparent funding, review, and corrective action.

Week 10 Assignment Briefing

Sustainability and Social-Impact Plan (maximum 1,200 words plus KPI appendix)

- **Due:** Beginning of Week 11 (group assignment)
- **Requirements:**
 1. define the principal noise, privacy, climate, and distributional impacts;
 2. propose a mitigation hierarchy: avoid, reduce, manage, and compensate residual impacts;
 3. calculate operational electricity emissions and one life-cycle indicator with sourced factors;
 4. prepare a privacy-governance plan covering necessity, minimization, security, retention, notice, and incidents;
 5. evaluate one equity or public-service commitment, including cost, beneficiary, and payer; and
 6. define at least five KPIs with boundaries, baselines, targets, measurement methods, owners, and review triggers.
- **Grading focus:** Economic reasoning (25%), environmental analysis (25%), privacy and governance (20%), equity and stakeholder analysis (20%), professionalism (10%).

Synthesis & Capstone Group Presentations

Preview question for next week:

How will you combine utility formulas, unit economics, infrastructure constraints, and sustainability into one master presentation that secures \$100M in venture funding?

Questions?

(Final 15 minutes reserved for extended Q&A)